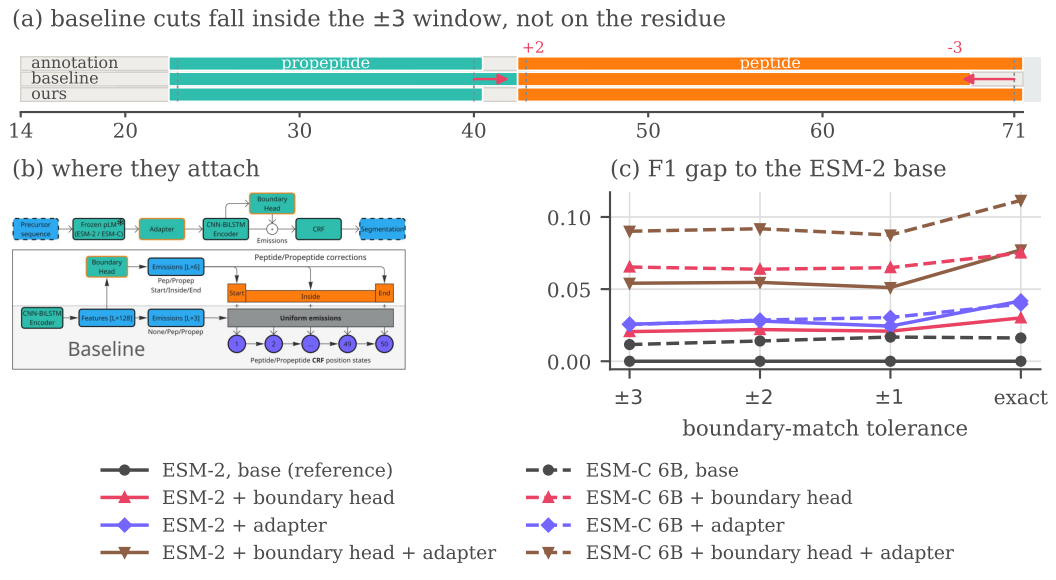
(a) baseline cuts fall inside the ±3 window, not on the residue
annotation
baseline
ours
propeptide
peptide
+2
-3
14
20
30
40
50
60
71
(b) where they attach
Precursor sequence
Frozen pLM (ESM-2 / ESM-C)
Adapter
CNN-BiLSTM Encoder
Boundary Head
Emissions
CRF
Segmentation
Boundary Head
Emissions [L×E]
Pep/Propep Start/Inside/End
Peptide/Propeptide corrections
Start
Inside
End
CNN-BiLSTM Encoder
Features [L×128]
Emissions [L×3]
None/Pep/Propep
Uniform emissions
1
2
3
40
50
Baseline
Peptide/Propeptide CRF position states
(c) F1 gap to the ESM-2 base
0.10
0.05
0.00
±3
±2
±1
exact
boundary-match tolerance
ESM-2, base (reference)
ESM-2 + boundary head
ESM-2 + adapter
ESM-2 + boundary head + adapter
ESM-C 6B, base
ESM-C 6B + boundary head
ESM-C 6B + adapter
ESM-C 6B + boundary head + adapter